# Smart Panlee

## Smart Serial LCD Display

## ZX3D95CE01S-TR-4848



## Features:

1. Based on Wireless-Tag's WT32-S3-WROVER-N16R8（16MB Flash/8Line 8M die inside）
2. 480*480 high-resolution RGB display with capacitive touch
3. 86box product appearance, convenient for embedding in smart home applications
4. Support online rapid prototyping on the 8MS development platform
5. Onboard RS485 chip

## Core Materials (Tab. 0):

| No. | Name | Model | Remark |
|---|---|---|---|
| 1 | ESP32-S3 module | WT32-S3-WROVER-N16R8 | |
| 2 | | | |
| 3 | | | |

## Naming instructions ：

| Specifications | illustrate | Remark |
|---|---|---|
| ZX3D95CE01S-TR | Contains temperature and humidity sensor and RS485, using on-board antenna | |
| ZX3D95CE01S-R | RS485 only, use onboard antenna | |
| ZX3D95CE01S-i-TR | Including temperature and humidity sensor and RS485, use IPEX external antenna | Batch to order |
| ZX3D95CE01S-i-R | RS485 only, use IPEX external antenna | Batch to order |

## Hardware Interface:

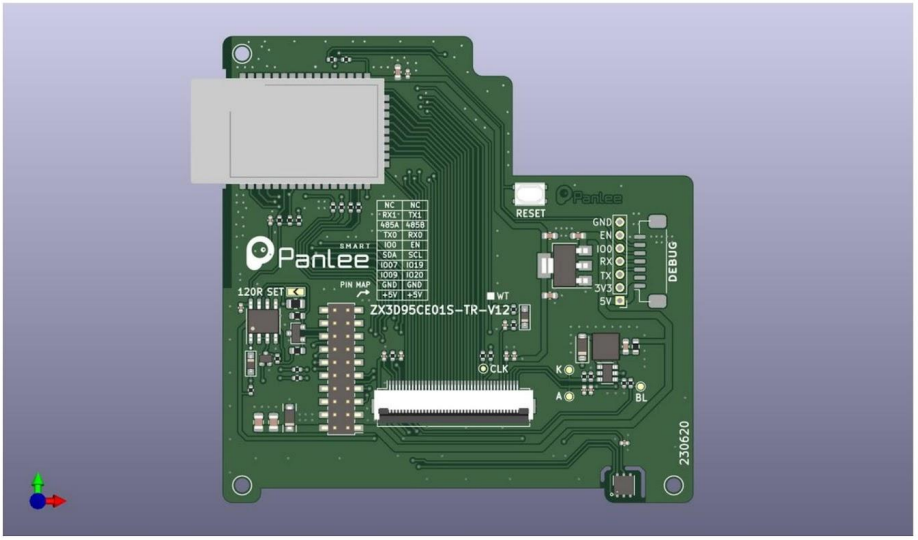



Fig.1 Hardware Interface

## Interface Description:

[1] Debugging Interface(Tab.1)

| Pin | Description | Module Pin | Voltage Range | Remark |
|---|---|---|---|---|
| 1 | +5V | - | 5V | |
| 2 | +3.3V | - | 3.3V | |
| 3 | TXD0 | TXD0 | 3.3V TTL | Burning serial port |
| 4 | RXD0 | RXD0 | 3.3V TTL | Burning serial port |
| 5 | EN | CHIP_PU | 3.3V MAX | Chip reset |
| 6 | IO0 | GPIO 0 | 3.3V TTL | Boot mode |
| 7 | GND | | | Selection Grounding |

[2] Sub Board Interface (Tab.2)

| Pin | Description | Module Pin | Voltage Range | Remark |
|---|---|---|---|---|
| 1、2 | +5V | - | +5V±5% | Board power supply |
| 3、4 | GND | - | 0V | Grounding |
| 5 | EXT_IO2 | GPIO 20 | 0~3.3V | Output interface 2 |
| 6 | EXT_IO1 | GPIO 9 | 0-3.3V | Output interface 1 |
| 7 | EXT_IO3 | GPIO 19 | 0-3.3V | Output interface 3 |
| 8 | EXT_IO4 | GPIO 7 | 0-3.3V | Output interface 4 |
| 9 | IIC_SCL | GPIO 6 | 3.3V TTL | IIC bus clock, multiplexed with touch screen |
| 10 | IIC_SDA | GPIO 15 | 3.3V TTL | IIC bus data, multiplexed with touch screen |
| 11 | EN | EN | 0~VCC | Chip enable |
| 12 | GPIO 0 | GPIO 0 | 0~VCC | BOOT selection |
| 13 | RXD | RXD0 | 3.3V TTL | Download debugging serial port |
| 14 | TXD | TXD0 | 3.3V TTL | Download debugging serial port |
| 15 | 485 B | - | RS485 | 485 bus negative, not available when output interface is used |
| 16 | 485 A | - | RS485 | 485 bus positive, not available when output interface is used |
| 17 | TXD_EXT | GPIO 2 | 3.3V TTL | Multiplexed with 485 serial port |
| 18 | RXD_EXT | GPIO 1 | 3.3V TTL | Multiplexed with 485 serial port |
| 19、20 | NC | - | - | not connect |

[3] Display Interface (Tab.3)

| Description | Module Pin | Remark |
|---|---|---|
| TP_RST | - | Using RC reset |
| TP_SCL | GPIO 6 | Touch IIC bus clock, multiplexed with external interface |
| TP_SDA | GPIO 15 | Touch IIC bus data, multiplexed with external interface |
| TP_INT | GPIO 4 | Touch interrupt |
| LCD_RST | - | LCD auto reset, use RGB_VS signal to keep working |
| LCD_CS | GPIO 38 | LCD SPI bus CS |
| LCD_SCLK | GPIO 45 | LCD SPI bus SCLK |
| LCD_MOSI | GPIO 48 | LCD SPI bus MOSI |
| RGB_PCLK | GPIO 39 | LCD RGB INTERFACE PCLK |
| RGB_DE | GPIO 40 | LCD RGB INTERFACE DE |
| RGB_VS | GPIO 41 | LCD RGB INTERFACE VS |
| RGB_HS | GPIO 42 | LCD RGB INTERFACE HS |
| RGB_D0 | GPIO 45 | LCD RGB INTERFACE D0 |
| RGB_D1 | GPIO 48 | LCD RGB INTERFACE D1 |
| RGB_D2 | GPIO 47 | LCD RGB INTERFACE D2 |
| RGB_D3 | GPIO 0 | LCD RGB INTERFACE D3 |
| RGB_D4 | GPIO 21 | LCD RGB INTERFACE D4 |
| RGB_D5 | GPIO 14 | LCD RGB INTERFACE D5 |
| RGB_D6 | GPIO 13 | LCD RGB INTERFACE D6 |
| RGB_D7 | GPIO 12 | LCD RGB INTERFACE D7 |
| RGB_D8 | GPIO 11 | LCD RGB INTERFACE D8 |
| RGB_D9 | GPIO 16 | LCD RGB INTERFACE D9 |
| RGB_D10 | GPIO 17 | LCD RGB INTERFACE D10 |
| RGB_D11 | GPIO 18 | LCD RGB INTERFACE D11 |

| | | |
|---|---|---|
| RGB_D12 | GPIO 8 | LCD RGB INTERFACE D12 |
| RGB_D13 | GPIO 3 | LCD RGB INTERFACE D13 |
| RGB_D14 | GPIO 46 | LCD RGB INTERFACE D14 |
| RGB_D15 | GPIO 10 | LCD RGB INTERFACE D15 |
| LCD_BL | GPIO 4 | LCD backlight control, high level enable |

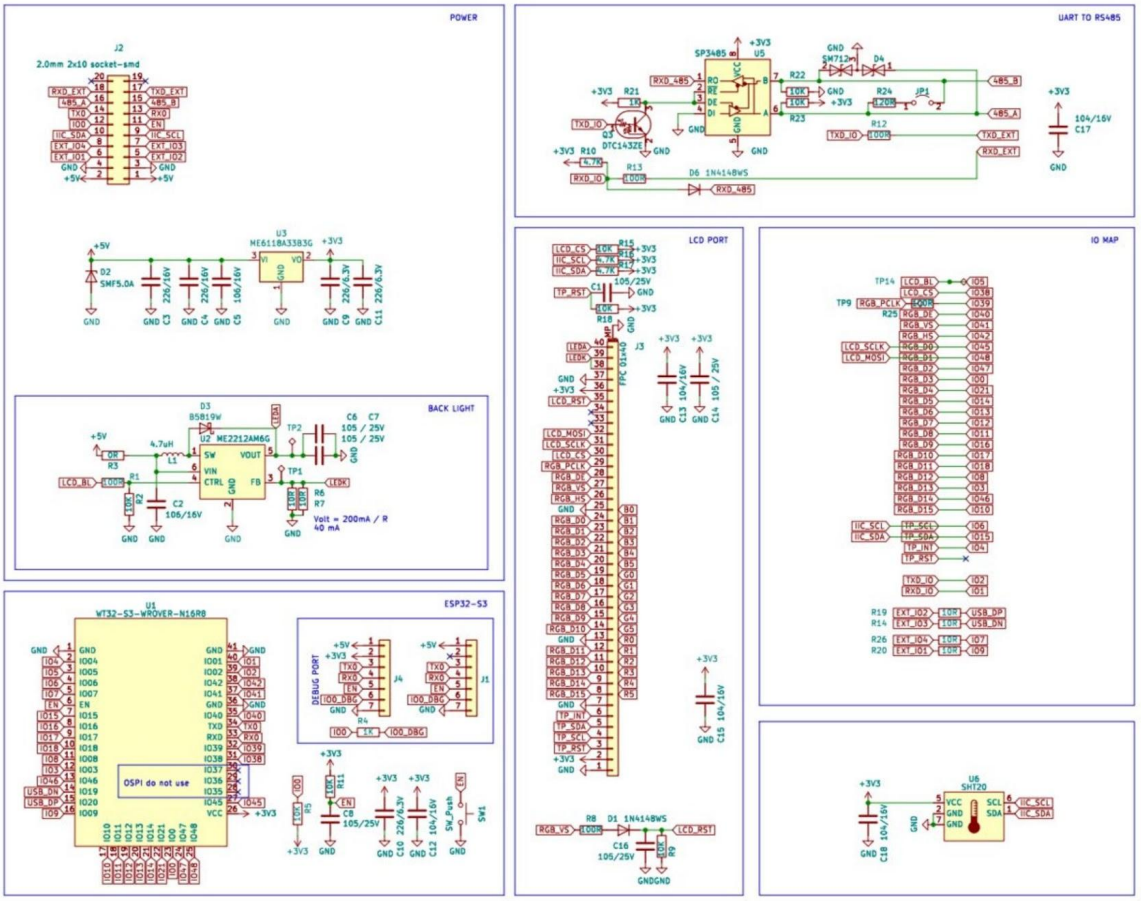
## Interface Encapsulation (Tab.4):

| Peripheral Name | Description |
|---|---|
| Debugging interface | Debugging interface MX1.25-7P |
| External interface | External interface 2.0mm 2x10P female header |

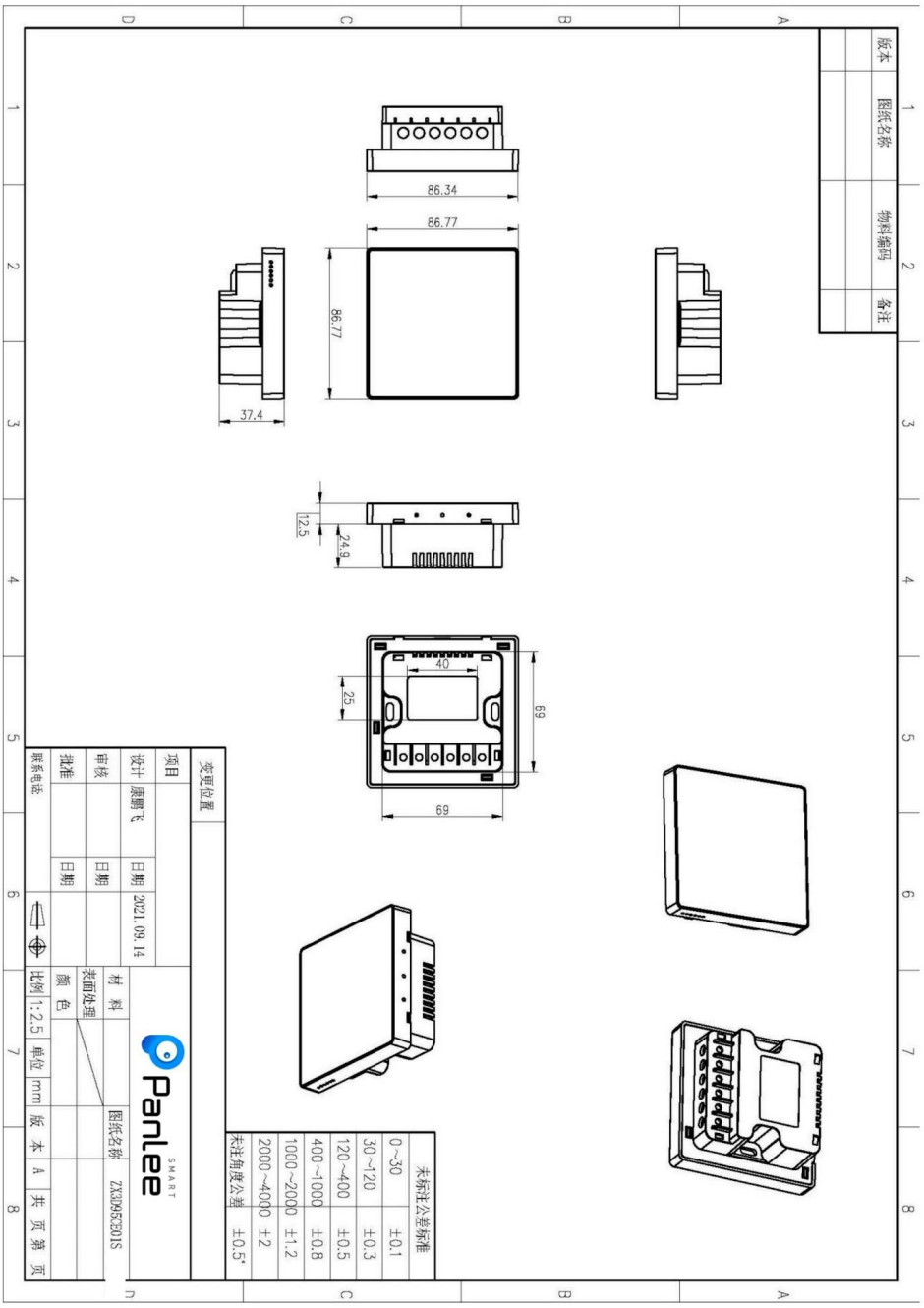
## Hardware peripherals：

| Appearance Name | Description |
|---|---|
| SHT20 | Temperature and humidity sensor, hanging on IIC bus |
| LCD | 480*480 resolution, RGB565 |
| RS485 | Use automatic transceiver 485 circuit |

## Schematic:


POWER
UART TO RS485
LCD PORT
IO MAP
BACK LIGHT
ESP32-S3
WT32-S3-WROVER-N16R8
OSPI do not use
DEBUG PORT
ME2212AM6G
SP3485
SHT20

## Outline Dimensional Drawing (Fig.2)



## Specification Parameters:

[1] Display Parameters (Tab.6)

| | |
|---|---|
| **LCD panel size** | 3.95inch |
| **Display type** | IPS |
| **Drive IC model** | GC9503V |
| **Visual angle** | FREE |
| **Physical dimensions** | 84.00*84.00*3.13mm |
| **Resolution** | 480(H)x3(RGB)x480(V) |
| **Interface** | RGB |
| **Color** | 16.7M |
| **Backlit mode** | 8 white LEDs in 4S2P |
| **Backlight brightness** | |

[2] Touch Parameters (Tab.7)

| | |
|---|---|
| **Touchscreen type** | Capacitive touch |
| **Drive IC model** | FT6336U |
| **Interface** | I2C |
| **Touchscreen structure** | GF |
| **Touch mode** | Surface touch |
| **Surface hardness** | 6H |
| **Light transmittance** | 85% |

## Firmware Burning:

1. Connect the downloader (ZXACC-ESPDB) via a USB-Type C cable. And then connect the ZX3D95CE01S-TR board with the downloader (ZXACC-ESPDB) through an cable. As the downloader (ZXACC-ESPDB) has automatic data flow processing capabilities, the firmware can be downloaded automatically through the ESP32 Flash Download Tools.



Fig.3

2. As shown in Fig. 4: Select the firmware path at mark 1, and then fill in the burning address, usually 0X00. Note that this checkbox must be checked; Set the crystal frequency to be 40MHz at mark 2; Select 32Mbit for Flash size at mark 3; Select DIO for SPI MODE at mark 4; Select the COM port number recognized by the computer at mark 5; Select the baud rate at mark 6 (the higher the value is, the faster the firmware will be downloaded. Max. 1152000bps).
3. After the previous configuration, click START at mark 7 to start burning the firmware.

Complete the above steps, and then press the reset button on the back of the development board to start running the firmware you just burned.

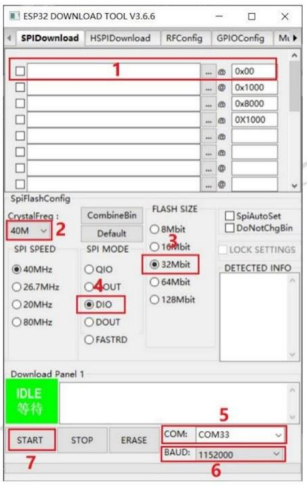



Fig.4

## Online GUI Designer:

Users can use our online GUI designer platform, which is similar to MIT APP Inventor, to realize the rapid GUI development with building blocks. Currently, the platform has perfected the graphic interface development, and more driver code blocks will be further improved in the future.

Login Page: http://8ms.xyz/login
User Manual: https://doc.smartpanle.cn/ESP32-S3/index.html

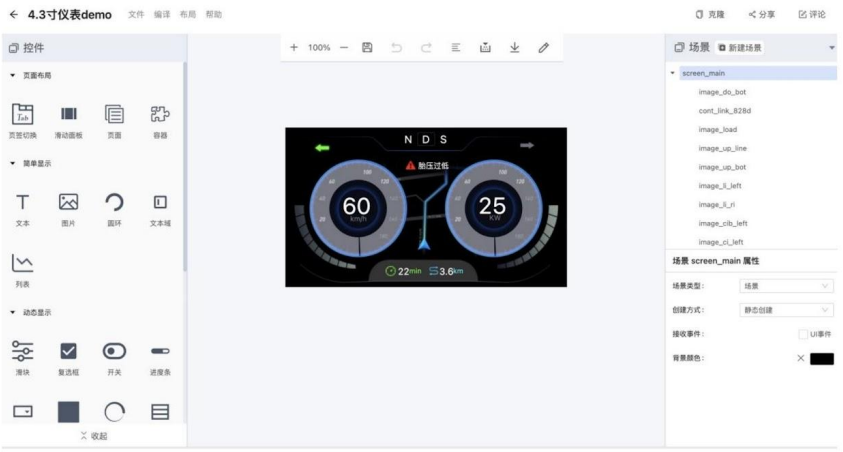



Fig.5

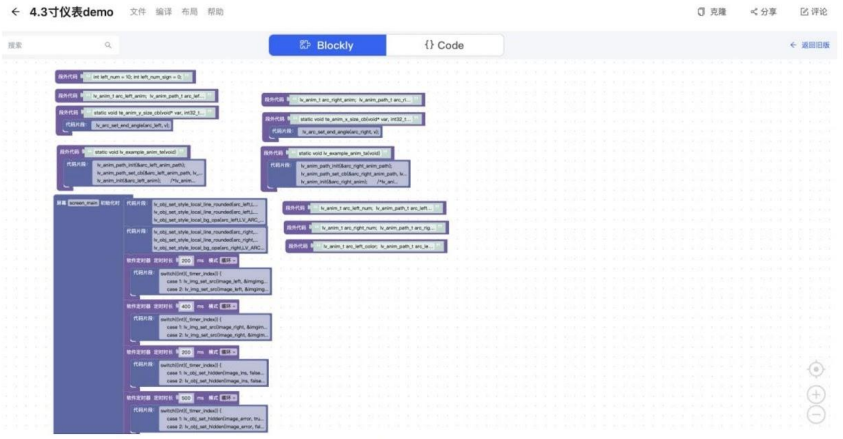


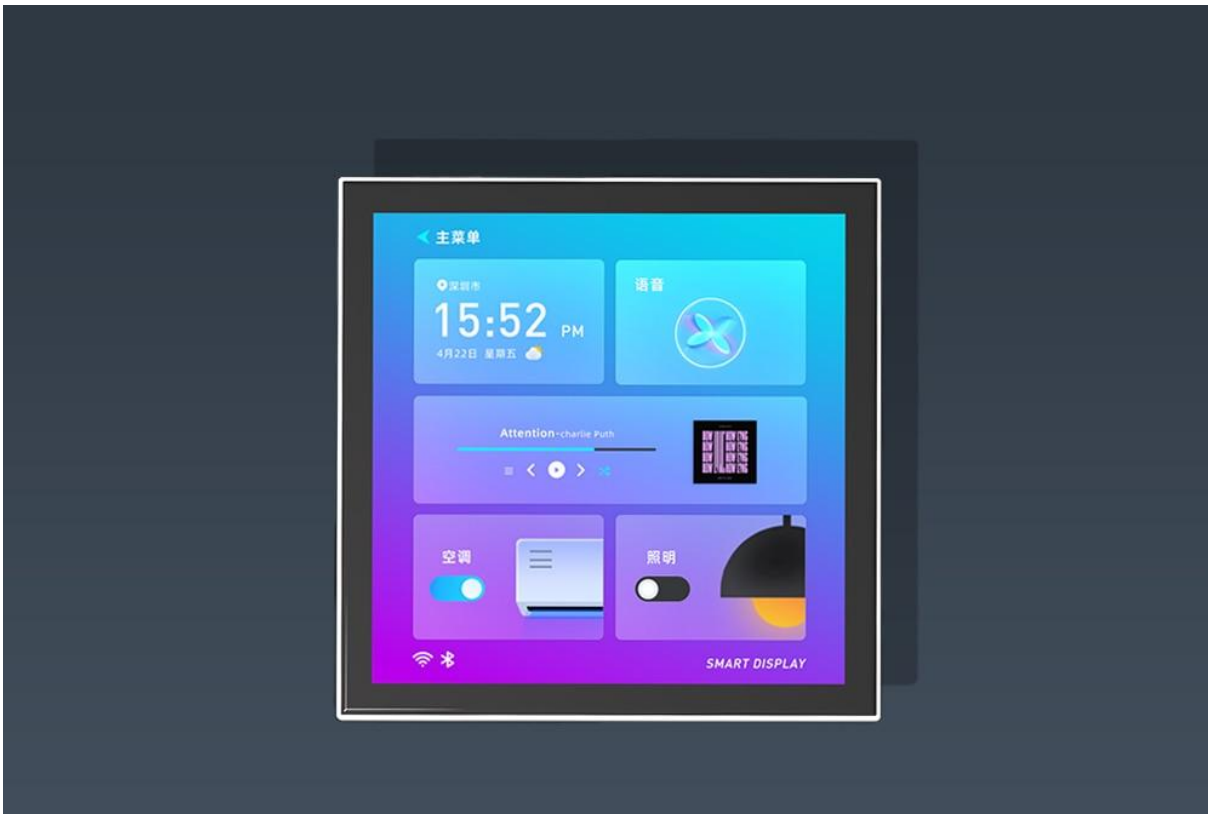

Fig.6

Please note:

Dear customer, this product requires software development, not directly connected to the power supply can be used, before purchasing, please know, whether it will be software development/operation, this product also requires a downloader/development board, Dupont cable, debugging the required burning tools need to be purchased separately (debugger this shop is available).

New products haven't been updated yet

The following pictures are for reference only



Parameter introduction:

- Module Model:WT32-S3-WROVER-N16R8
- Screen size:4 inch IPS full screen
- Resolution:480*480,RGB565
- Screen driver IC: GC9503V
- Touch Driver IC:FT6336U
- Sensor:Temperature and humidity sensing, SHT20
- RS485: using automatic transceiver 485 circuitry

## Hardware Interface:

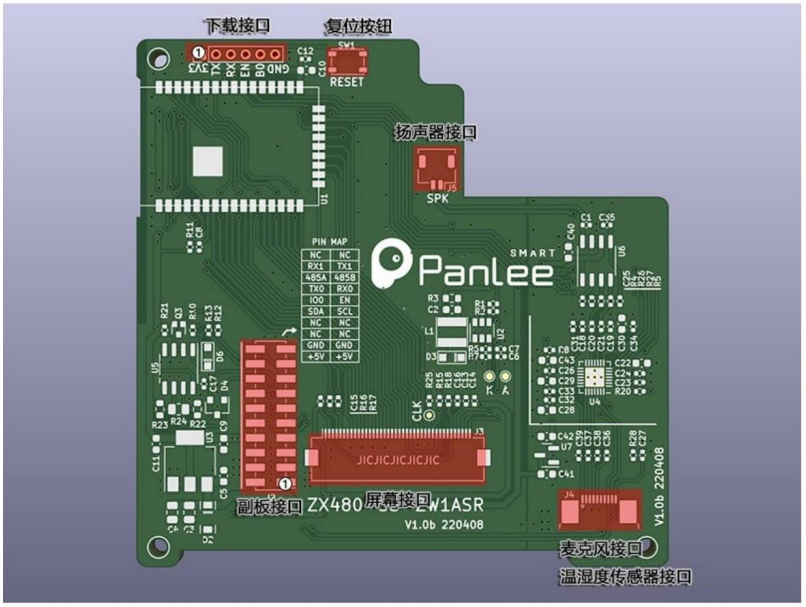



Fig.1 Hardware Interface

## Interface Description:

[1] Download Interface(Tab.1)

| Pin | Description | Module Pin | Voltage Range | Remark |
|---|---|---|---|---|
| 1 | 3V3 | VCC | 3.3V | Module power supply |
| 2 | TXD | TXD0 | 3.3V TTL | Download and debug serial interface |
| 3 | RXD | RXD0 | 3.3V TTL | Download and debug serial interface |
| 4 | EN | EN | 0~VCC | Chip enable |
| 5 | BOOT | GPIO 0 | 0~VCC | BOOT select |
| 6 | GND | GND | 0V | Ground |

[2] Sub Board Interface (Tab.2)

| Pin | Description | Module Pin | Voltage Range | Remark |
|---|---|---|---|---|
| 1、2 | +5V | - | 5V±5% | Board power supply |
| 3、4 | GND | - | 0V | Ground |
| 5-8 | NC | - | - | should be left floating |
| 9 | IIC_SCL | GPIO 6 | 3.3V TTL | IIC bus, multiplexed with multiple peripherals |
| 10 | IIC_SDA | GPIO 7 | 3.3V TTL | IIC bus, multiplexed with multiple peripherals |
| 11 | EN | EN | 0~VCC | Chip enable |
| 12 | GPIO 0 | GPIO 0 | 0~VCC | BOOT select, multiplexed with LCD_CS |
| 13 | RXD | RXD0 | 3.3V TTL | Download and debug serial interface |
| 14 | TXD | TXD0 | 3.3V TTL | Download and debug serial interface |
| 15 | 485 B | - | RS485 | 485 bus negative, unavailable when an output interface is used |
| 16 | 485 A | - | RS485 | 485 bus positive, unavailable when an output interface is used |
| 17 | TXD_EXT | GPIO 4 | 3.3V TTL | Multiplexed with 485 serial interface |
| 18 | RXD_EXT | GPIO 5 | 3.3V TTL | Multiplexed with 485 serial interface |
| 19、20 | NC | - | - | not connect |

[3] Display Interface (Tab.3)

| Description | Module Pin | Remark |
|---|---|---|
| TP_RST | - | RC reset is used |
| TP_SCL | GPIO 6 | Touch panel IIC, multiplexed with multiple peripherals |
| TP_SDA | GPIO 7 | Touch panle IIC, multiplexed with multiple peripherals |
| TP_INT | - | Not connect |
| LCD_RST | - | RC reset is used |
| LCD_CS | GPIO 0 | LCD SPI bus CS |
| LCD_SCLK | GPIO 10 | LCD SPI bus SCLK |
| LCD_MOSI | GPIO 9 | LCD SPI bus MOSI |
| RGB_PCLK | GPIO 14 | LCD RGB interface PCLK |
| RGB_DE | GPIO 13 | LCD RGB interface DE |
| RGB_VS | GPIO 12 | LCD RGB interface VS |
| RGB_HS | GPIO 11 | LCD RGB interface HS |
| RGB_D0 | GPIO 10 | LCD RGB interface D0 |
| RGB_D1 | GPIO 9 | LCD RGB interface D1 |
| RGB_D2 | GPIO 40 | LCD RGB interface D2 |
| RGB_D3 | GPIO 20 | LCD RGB interface D3 |
| RGB_D4 | GPIO 19 | LCD RGB interface D4 |
| RGB_D5 | GPIO 41 | LCD RGB interface D5 |
| RGB_D6 | GPIO 46 | LCD RGB interface D6 |
| RGB_D7 | GPIO 3 | LCD RGB interface D7 |
| RGB_D8 | GPIO 42 | LCD RGB interface D8 |
| RGB_D9 | GPIO 8 | LCD RGB interface D9 |
| RGB_D10 | GPIO 18 | LCD RGB interface D10 |
| RGB_D11 | GPIO 2 | LCD RGB interface D11 |

| | | |
|---|---|---|
| RGB_D12 | GPIO 17 | LCD RGB interface D12 |
| RGB_D13 | GPIO 16 | LCD RGB interface D13 |
| RGB_D14 | GPIO 1 | LCD RGB interface D14 |
| RGB_D15 | GPIO 15 | LCD RGB interface D15 |
| LCD_BL | GPIO 45 | LCD backlight control, active high |
| RGB_D7 | GPIO 3 | LCD RGB interface D7 |
| RGB_D8 | GPIO 42 | LCD RGB interface D8 |
| RGB_D9 | GPIO 8 | LCD RGB interface D9 |
| RGB_D10 | GPIO 18 | LCD RGB interface D10 |
| RGB_D11 | GPIO 2 | LCD RGB interface D11 |
| RGB_D12 | GPIO 17 | LCD RGB interface D12 |
| RGB_D13 | GPIO 16 | LCD RGB interface D13 |
| RGB_D14 | GPIO 1 | LCD RGB interface D14 |
| RGB_D15 | GPIO 15 | LCD RGB interface D15 |
| LCD_BL | GPIO 45 | LCD backlight control, active high |

[4] Microphone Connector/Temperature and Humidity Sensor Interface

Connect dual microphones and IIC temperature and humidity sensor by using an expansion board.

## Hardware Peripherals(Tab.4):

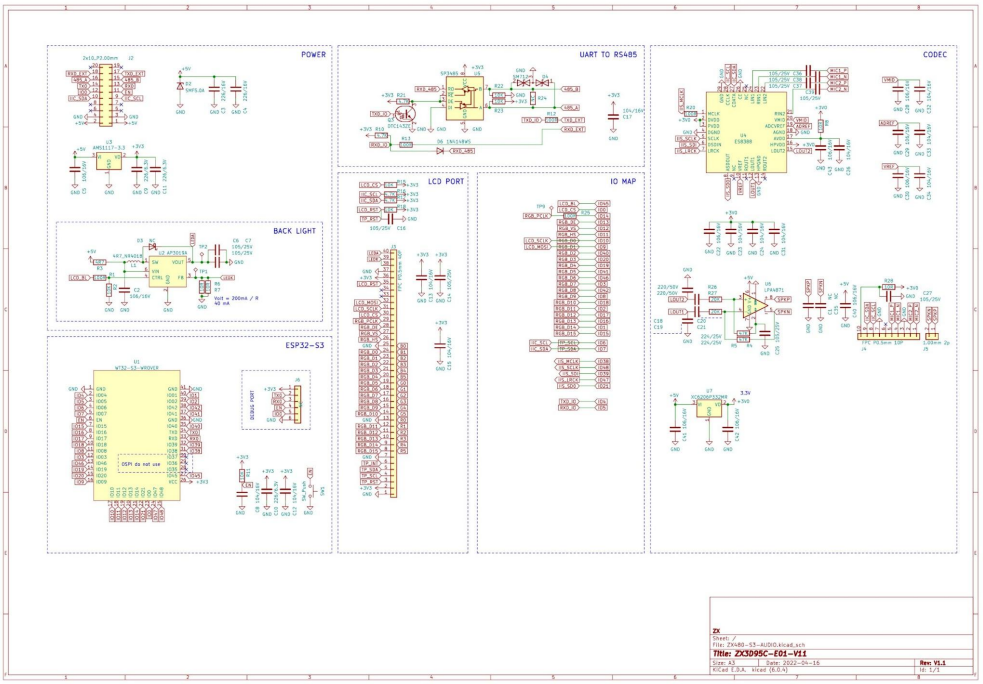
| Peripheral Name | Description |
|---|---|
| RS485 | 485 module |
| LCD | LCD with RGB interface |

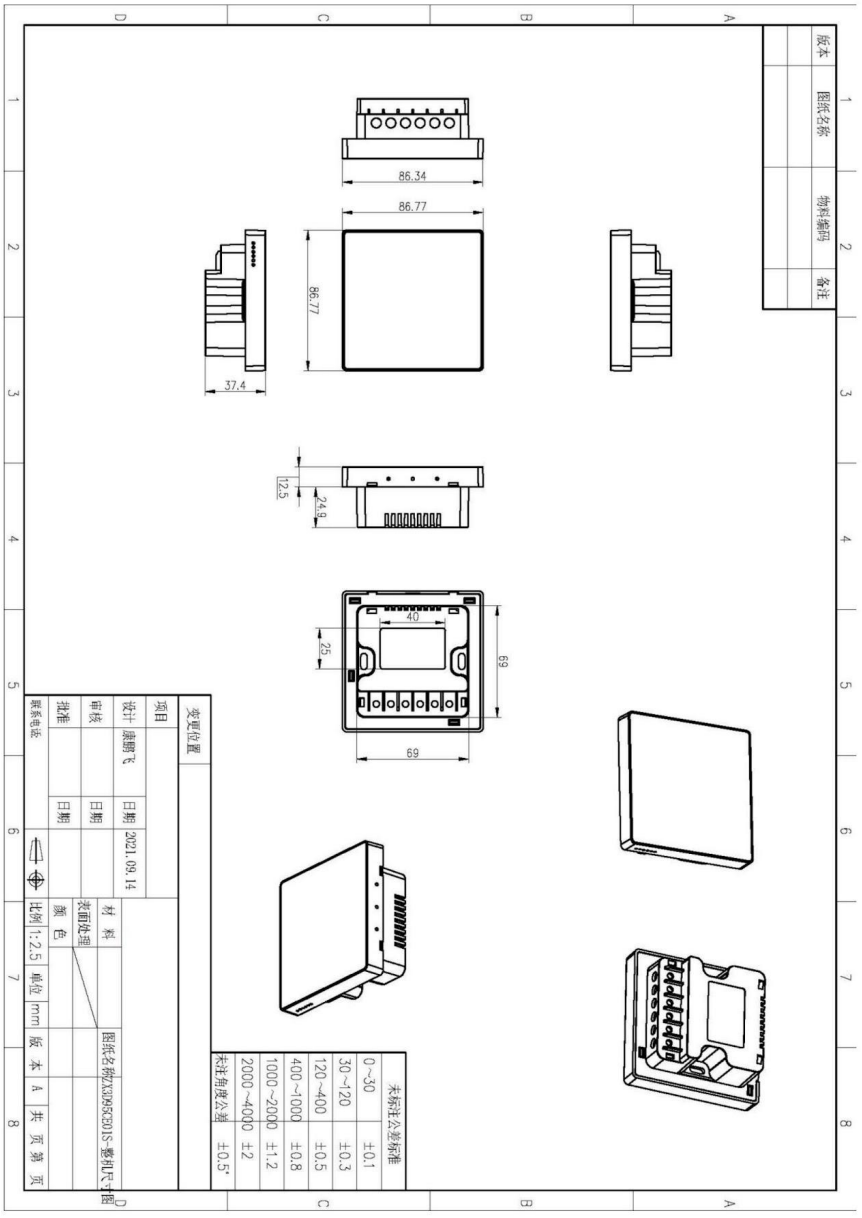
[1] CODEC Definition (Tab.5)

| Description | Module Pin | Remark |
|---|---|---|
| IIS_MCLK | GPIO 38 | IIS bus MCLK |

| | | |
|---|---|---|
| IIS_SCLK | GPIO 48 | IIS bus SCLK |
| IIS_LRCK | GPIO 47 | IIS bus LRCK |
| IIS_SDO | GPIO 21 | IIS bus SDO |
| IIS_SDI | GPIO 39 | IIS bus SDI |
| IIC_SCL | GPIO 6 | IIC bus, used to configure CODEC, multiplexed with multiple peripherals |
| IIC_SDA | GPIO 7 | IIC bus, used to configure CODEC, multiplexed with multiple peripherals |

## Schematic:


POWER
UART TO RS485
CODEC
BACK LIGHT
LCD PORT
IO MAP
ESP32-S3
ZX
Sheet: /
File: ZX480-S3-AUDIO.kicad_sch
Title: ZX3095C-E01-V11
Size: A3
Date: 2022-04-16
Rev: V1.1
KiCad E.D.A. kicad (6.0.6)
Id: 1/1


86.34
86.77
86.77
37.4
12.5
24.9
40
25
69
69
版本
图纸名称
物料编码
备注
变更位置
项目
设计 康鹏飞 日期 2021.09.14
审核 日期
批准 日期
联系电话
比例 1:2.5
单位 mm
版本 A
共 页 第 页
材料
表面处理
颜色
图纸名称ZX3D95CE01S-整机尺寸图
未标注公差标准
0~30 ±0.1
30~120 ±0.3
120~400 ±0.5
400~1000 ±0.8
1000~2000 ±1.2
2000~4000 ±2
未注角度公差 ±0.5°

## Firmware Burning:

1. Connect the downloader (ZXACC-ESPDB) via a USB-Type C cable. And then connect the ZX3D50CE02S board with the downloader (ZXACC-ESPDB) through an cable. As the downloader (ZXACC-ESPDB) has automatic data flow processing capabilities, the firmware can be downloaded automatically through the ESP32 Flash Download Tools.



Fig.3

2. As shown in Fig. 4: Select the firmware path at mark 1, and then fill in the burning address, usually 0X00. Note that this checkbox must be checked; Set the crystal frequency to be 40MHz at mark 2; Select 32Mbit for Flash size at mark 3; Select DIO for SPI MODE at mark 4; Select the COM port number recognized by the computer at mark 5; Select the baud rate at mark 6 (the higher the value is, the faster the firmware will be downloaded. Max. 1152000bps).

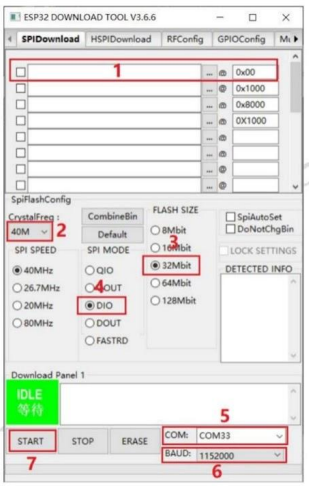



Fig.4

3. After the previous configuration, click START at mark 7 to start burning the firmware.

4. Complete the above steps, and then press the reset button on the back of the development board to start running the firmware you just burned.

ZX3D95CEO1S-TR and ZX3D95CEO1S-R, we only provide semi-finished products, the chassis is left empty; users can add relay or RS485 power supply module according to the actual needs of their products.

